

A diagonal counterexample to Besov spectral mapping

A negative answer to the final question in arXiv:1910.06369, Section 8.2

Automated mathematics discovery run `fa_banach_001`
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Status: candidate counterexample, likely valid. For every $t > 0$ the bounded diagonal operator $f(tA)$ has an extra spectral limit point which is absent from $f(\sigma(A)) \cup \{0\}$.

1 Source question

Batty, Gomilko, and Tomilov define an analytic Besov algebra $\mathcal{B}$ and its functional calculus for generators of bounded semigroups [1]. Their Theorem 7.3 proves, under the additional hypothesis that f vanishes at infinity,

$$\sigma(f(A)) \cup \{0\} = f(\sigma(A)) \cup \{0\}.$$

The exact formula in the source is reproduced here.

Theorem 7.3. *Let $(e^{-tA})_{t \geq 0}$ be a bounded C_0 -semigroup on a Hilbert space X . If $f \in \mathcal{B} \cap C_0(\overline{\mathbb{C}}_+)$, then*

$$(7.2) \quad \sigma(f(A)) \cup \{0\} = f(\sigma(A)) \cup \{0\}.$$

In particular, if $\mu \in M_0(\mathbb{R}_+)$ then

$$\sigma(\mathcal{L}\mu(A)) \cup \{0\} = \mathcal{L}\mu(\sigma(A)) \cup \{0\}.$$

On page 43 they ask whether the equality remains true for $f(tA)$ if $f \in \mathcal{B}$ and $t \mapsto f(tA)$ is norm-continuous on $(0, \infty)$.

Open question. Let us assume that A and f are as in Theorem 8.8 so that $f(\cdot A) \in C((0, \infty), L(X))$. If (8.8) holds and $f(z) = e^{-z}$, then $(f(tA))_{t \geq 0}$ is an immediately norm-continuous C_0 -semigroup, and it is well-known that the spectral mapping theorem then holds in the form of (7.2). We raise the question:

If $f \in \mathcal{B}$ and $f(\cdot A) \in C((0, \infty), L(X))$, does (7.2) hold for $f(tA)$, $t > 0$?

We answer this final explicitly stated question negatively, in the simplest normal diagonal setting.

2 Statement and proof intuition

Recall the source definition

$$\mathcal{B} = \left\{ f \in \text{Hol}(\mathbb{C}_+) : \int_0^\infty \sup_{\beta \in \mathbb{R}} |f'(\alpha + i\beta)| d\alpha < \infty \right\},$$

with norm given by the displayed seminorm plus $\|f\|_\infty$.

Theorem 1. *There exist a bounded C_0 -semigroup with negative generator $-A$ on a Hilbert space and $f \in \mathcal{B}$ such that*

$$t \mapsto f(tA) \quad \text{belongs to } C((0, \infty), \mathcal{L}(X)),$$

but, for every $t > 0$,

$$\sigma(f(tA)) \cup \{0\} \neq f(t\sigma(A)) \cup \{0\}.$$

Proof intuition. Choose an unbounded discrete spectrum and a Besov function with a nonzero limit at infinity. The values of f on the discrete spectrum converge to that limit. Hence the spectrum of the resulting bounded diagonal operator, which is the closure of its diagonal entries, acquires one extra point. The exact rational formula makes the dilation family norm-continuous away from zero. Thus norm continuity does not repair the missing “point at infinity” in the proposed spectral image.

3 Proof

Let $X = \ell_2(\mathbb{N})$, with canonical basis $(e_n)_{n \geq 1}$, and define

$$Ae_n = ine_n, \quad D(A) = \left\{ x = (x_n) : \sum_{n \geq 1} n^2 |x_n|^2 < \infty \right\}.$$

Then A is closed and densely defined, and

$$T(s)e_n = e^{-ins}e_n \quad (s \geq 0)$$

is a unitary C_0 -semigroup with generator $-A$. Strong continuity follows immediately from dominated convergence applied to $\sum_n |e^{-ins} - 1|^2 |x_n|^2$.

Take

$$f(z) = \frac{z}{1+z} = 1 - \frac{1}{1+z}.$$

For $z = \alpha + i\beta$ in $\mathbb{C}_+$,

$$f'(z) = \frac{1}{(1+z)^2}, \quad \sup_{\beta \in \mathbb{R}} |f'(\alpha + i\beta)| = (1+\alpha)^{-2}.$$

Consequently

$$\int_0^\infty \sup_{\beta} |f'(\alpha + i\beta)| d\alpha = 1.$$

Moreover $|z| < |1+z|$ on $\mathbb{C}_+$, so $\|f\|_\infty = 1$. Thus $f \in \mathcal{B}$, and its value at infinity is $f(\infty) = 1$.

Compatibility of the $\mathcal{B}$ -calculus with the resolvent calculus [1, Theorem 1.1(b)] gives, for $t > 0$,

$$f(tA) = I - (I + tA)^{-1}, \quad f(tA)e_n = b_n(t)e_n, \quad b_n(t) := \frac{itn}{1 + itn}.$$

If $s, t > 0$, then

$$\begin{aligned} |b_n(t) - b_n(s)| &= \frac{|t - s|n}{\sqrt{1 + t^2 n^2} \sqrt{1 + s^2 n^2}} \\ &\leq \frac{|t - s|}{\min\{s, t\}}. \end{aligned}$$

The operator norm of a bounded diagonal operator is the supremum of the moduli of its entries. Taking the supremum over n therefore proves that $t \mapsto f(tA)$ is norm-continuous on $(0, \infty)$.

The spectrum of the unbounded diagonal operator is

$$\sigma(A) = \{in : n \geq 1\}.$$

Indeed, away from this closed discrete set the diagonal inverse of $A - \lambda$ is bounded. A bounded diagonal operator has spectrum equal to the closure of its diagonal entries: outside that closure the coordinatewise inverse is bounded, while points in the closure are eigenvalues or approximate eigenvalues. Since $b_n(t) \rightarrow 1$,

$$\sigma(f(tA)) = \{b_n(t) : n \geq 1\} \cup \{1\}.$$

In particular e_n is an approximate eigenvector sequence at 1, because

$$\|(f(tA) - I)e_n\| = |b_n(t) - 1| = \frac{1}{\sqrt{1 + t^2 n^2}} \rightarrow 0.$$

On the other hand,

$$f(t\sigma(A)) \cup \{0\} = \{b_n(t) : n \geq 1\} \cup \{0\}.$$

The point 1 does not occur on the right: $b_n(t) = 1$ would imply $itn = 1 + itn$. Thus the two sets differ for every $t > 0$, proving Theorem 1. $\square$

4 Scope, verification, and novelty

The mechanism is exactly excluded by the positive Theorem 7.3 in the source: that theorem assumes $f \in \mathcal{B} \cap C_0(\overline{\mathbb{C}}_+)$, whereas here $f(\infty) = 1$. Adding $\{0\}$ cannot conceal the nonzero extra point.

There is one interpretive caveat. The paragraph containing the question first refers to Theorem 8.8, where a resolvent-decay hypothesis is a sufficient condition for norm continuity, and then asks the final question using only $f \in \mathcal{B}$ and norm continuity. The example answers that final formulation. It does not satisfy the stronger decay condition: for fixed $\alpha > 0$ and $\beta = n$, the relevant diagonal resolvent has norm $1/\alpha$, while $|f(\alpha + in)| \rightarrow 1$. No claim is made about a strengthened reading that retains that extra condition.

The included deterministic checker verifies the scalar identities, continuity bound, and convergence to the extra point. The proof itself is exact and does not depend on computation.

The run indexes were searched by arXiv id, title, and core terms. A bounded primary arXiv search used the exact question, source title, authors, and related Besov-calculus papers; it found no explicit answer or this diagonal example. This is not an exhaustive novelty certification. We recommend human review, with priority given to the intended scope of the opening Theorem 8.8 reference.

References

- [1] C. Batty, A. Gomilko, and Yu. Tomilov, *The theory of Besov functional calculus: developments and applications to semigroups*, arXiv:1910.06369 (2019; revised 2021).